① # Basic operations and notations

t = time variable

X_t = time series (observations at time 't')

$\{X_t\}$ = stochastic process (any function)

X_{t-k} = k-lagged series (X_{t-1} = 1-step lag series)

X_{t+k} = k-lead series (X_{t+1} = 1-step lead series)

t	X_t	X_{t-1}	X_{t-2}	X_{t+1}	X_{t+2}
1	x_1	—	—	x_2	x_3
2	x_2	x_1	—	x_3	x_4
3	x_3	x_2	x_1	x_4	x_5
4	x_4	x_3	x_2	x_5	x_6
5	x_5	x_4	x_3	x_6	x_7
6	x_6	x_5	x_4	x_7	x_8
⋮	⋮	⋮	⋮	⋮	⋮
$n-1$	x_{n-1}	x_{n-2}	x_{n-3}	x_n	x_{n+1}
n	x_n	x_{n-1}	x_{n-2}	—	—

Some usefull operators: ②

1) Backward shift operator (B)

when it apply on the series it make the series lagged series

$$B X_t = X_{t-1} \quad , \quad B X_{t-1} = X_{t-2} = B^2 X_t$$

$$B^2 X_t = X_{t-2}$$

$$\vdots \qquad \vdots$$

$$B^m X_t = X_{t-m}$$

no. of steps of lagged depends on the power of B

2) Forward shift operator (F)

This operator apply the series it make the series lead series.

$$F X_t = X_{t+1}$$

$$F^2 X_t = X_{t+2}$$

$$\vdots$$

$$F^m X_t = X_{t+m}$$

Note:

$$B^{-1} = F$$

$$B^{-1} X_t = X_{t-(-1)} = X_{t+1} = F X_t$$

So, $B^{-1} = F$

And $F^{-1} = B$

3) Difference operator (∇)

$$\nabla X_t = X_t - X_{t-1}$$

$$\nabla^2 X_t = \nabla(\nabla X_t)$$

$$= \nabla(X_t - X_{t-1})$$

$$= \nabla X_t - \nabla X_{t-1}$$

$$= (X_t - X_{t-1}) - (X_{t-1} - X_{t-2})$$

$$= X_t - X_{t-1} - X_{t-1} + X_{t-2}$$

$$\nabla^2 X_t = X_t - 2X_{t-1} + X_{t-2}$$

Now,

$$\nabla X_t = X_t - X_{t-1}$$

$$= X_t - BX_t$$

$$= (1-B)X_t$$

$$\nabla = 1-B$$

$$\nabla^2 X_t = X_t - 2X_{t-1} + X_{t-2}$$

$$= X_t - 2BX_t + B^2X_t$$

$$= (1-2B+B^2)\,X_t$$

$$\nabla^2 X_t = (1-B)^2 X_t$$

So, $\nabla^2 = (1-B)^2$

Note

$$\nabla = 1-B$$

$$\nabla^2 = (1-B)^2$$

$$\nabla^3 X_t = (1-B)^3 X_t$$

$$= (1-3B+3B^2-B^3)\,X_t$$

$$= X_t - 3BX_t + 3B^2X_t - B^3X_t$$

$$\nabla^3 X_t = X_t - 3X_{t-1} + 3X_{t-2} - X_{t-3}$$

Time series depends on time not on variable, If time is Continous then series is Continous otherwise it is discrete.

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Next see at pg # 14 Stochastic Process

and do onward from the notes.
to page 24

Autocorrelation

A sample autocorrelation coefficients measure the correlation between observations at different distances apart. These coefficients often provide insight into the probability model which generated the data.

As we are familiar with the ordinary correlation coefficient, namely that given N pairs of observations on two variables x and y the correlation coefficient is given by

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

A similar idea can applied to time series to see if successive observations are correlated.

Given N observations $x_1, x_2, \text{----} x_N$ on a discrete time series we can form $N-1$ pairs of observations, namely $(x_1, x_2), (x_2, x_3)$ ----- (x_{N-1}, x_N). Regarding the first observation in each pair as one variable and the second observation as a second variable, the correlation coefficient between x_i and x_{t+1} is given by

$$r_1 = \frac{\sum_{t=1}^{N-1} (x_t - \bar{x}_1)(x_{t+1} - \bar{x}_2)}{\sqrt{\sum_{t=1}^{N-1} (x_t - \bar{x}_1)^2 \sum_{t=1}^{N-1} (x_{t+1} - \bar{x}_2)^2}} \quad \text{———} (1)$$

where

$$\bar{x}_1 = \frac{\sum_{t=1}^{N-1} x_t}{N-1}$$ is the mean

of the first N-1 observations.

$$\bar{x}_2 = \frac{\sum_{t=2}^{N} x_t}{N-1}$$

is the mean of the last N-1 observations.

As eq ① measures correlation between successive observations, it is called autocorrelation coefficient or serial coefficient.

As $\bar{x}_1 \cong \bar{x}_2$ it is usually approximated by

$$r_1 = \frac{\sum_{t=1}^{N-1} (x_t - \bar{x})(x_{t+1} - \bar{x}) / N-1}{\sum_{t=1}^{N} (x_t - \bar{x})^2 / N}$$

where

$$\bar{x} = \frac{\sum_{t=1}^{N} x_t}{N}$$ is the overall

mean. Some authors also drop the factor $\frac{N}{N-1}$, which is close to one for large N, to give the even simpler formula

$$r_1 = \frac{\sum_{t=1}^{N-1} (x_t - \bar{x})(x_{t+1} - \bar{x})}{\sum_{t=1}^{N} (x_t - \bar{x})^2}$$

In a similar way we can find the correlation between observations a distance k apart, which is given by

$$r_k = \frac{\sum_{t=1}^{N-k} (x_t - \bar{x})(x_{t+k} - \bar{x})}{\sum_{t=1}^{N} (x_t - \bar{x})^2}$$

This is called the autocorrelation Coefficient at lag k.

In practice the autocorrelation coeffients are usually calculated by computing the series of autocovariance coefficient $\{C_k\}$, which we define by analogy with the usual covariance formula as

$$C_k = \frac{\sum_{t=1}^{N-k} (x_t - \bar{x})(x_{t+k} - \bar{x})}{N}$$

This is the autocovariance coefficient at lag k. We then compute

$$r_k = \frac{C_k}{C_0} \quad \text{(ii)}$$

for $k = 1, 2, \ldots, m$ where $m < N$.

There is often little point in calculating r_k for values of k greater than about $\frac{N}{4}$

Note that some authors prefer to use

$$c_k = \frac{\sum_{t=1}^{N-k} (x_t - \bar{x})(x_{t+k} - \bar{x})}{N-k}$$

but there is little difference for large N.

$$\gamma_k = \text{Cov}\left[y_t , y_{t+k}\right]$$

$$= E(y_t - \mu)(y_{t+k} - \mu)$$

$$= t + k - t$$

$$\gamma_k = k$$

$$\rho_k = \frac{E\left[(y_t - \mu)(y_{t+k} - \mu)\right]}{\sqrt{E\left[(y_t - \mu)^2\right] E\left[(y_{t+k} - \mu)^2\right]}}$$

$$\rho_k = \frac{E\left[(y_t - \mu)(y_{t+k} - \mu)\right]}{\sigma_Y^2}$$

$$\rho_k = \frac{\gamma_k}{\gamma_0}$$

Example.

t	y_t	y_{t+1}	y_{t+2}	$(y_t-\bar{y})(y_{t+1}-\bar{y})$	$(y_t-\bar{y})^2$
1	13	8	15	-6	9
2	8	15	4	-10	4
3	15	4	4	-30	25
4	4	4	12	36	36
5	4	12	11	-12	36
6	12	11	7	2	4
7	11	7	14	-3	1
8	7	14	12	-12	9
9	14	12	—	8	16
10	12	—	—		
	100			-27	144

$$\bar{y}_t = \frac{\sum_{t=1}^{N} y_t}{N}$$

$$\bar{y}_t = \frac{100}{10} = 10$$

$$r_k = \frac{\sum_{t=1}^{N-k} (y_t - \bar{y})(y_{t+k} - \bar{y})}{\sum_{t=1}^{N} (y_t - \bar{y})^2}$$

put $k=1$

$$r_1 = \frac{\sum_{t=1}^{N-1} (y_t - \bar{y})(y_{t+1} - \bar{y})}{\sum_{t=1}^{N} (y_t - \bar{y})^2}$$

$$r_1 = \frac{-27}{144} = -0.188$$

put $k=2$

$$r_2 = \frac{\sum_{t=1}^{N-2} (y_t - \bar{y})(y_{t+2} - \bar{y})}{\sum_{t=1}^{N} (y_t - \bar{y})^2}$$

$(y_t - \bar{y})(y_{t+2} - \bar{y})$

15
12
-30
-12
-6
-6
4
-6

-29

$$r_2 = \frac{-29}{144}$$

$$= -0.2014$$

K کی ویلیو تب تک
بڑھائیں جب تک
$\frac{N}{4}$ سے چھوٹی رہتی ہے۔

put $k = 3$

we get

$$r_3 = \frac{26}{144} = 0.181$$

As we increase lag k r_k becomes positive

Now,

t	y_t	y_{t-1}	y_{t-2}	$(y_t-\bar{y})(y_{t-1}-\bar{y})$	$(y_t-\bar{y})(y_{t-2}-\bar{y})$
1	13	—	—	—	—
2	8	13	—	−6	—
3	15	8	13	−10	15
4	4	15	8	−30	12
5	4	4	15	36	−30
6	12	4	4	−12	−12
7	11	12	4	2	−6
8	7	11	12	−3	−6
9	14	7	11	−12	4
10	12	14	7	8	−6
	100			−27	−29

$$r_{-k} = \frac{\sum_{t=k+1}^{N} (y_t - \bar{y})(y_{t-k} - \bar{y})}{\sum_{t=1}^{N} (y_t - \bar{y})^2}$$

put $k = +1$

$$r_{-1} = \frac{\sum_{t=2}^{N} (y_t - \bar{y})(y_{t-1} - \bar{y})}{\sum_{t=1}^{N} (y_t - \bar{y})^2}$$

$$= \frac{-27}{144} = -0.1875$$

put $k = 2$

$$r_{-2} = \frac{\sum_{t=3}^{N} (y_t - \bar{y})(y_{t-2} - \bar{y})}{\sum_{t=1}^{N} (y_t - \bar{y})^2}$$

$$r_{-2} = \frac{-29}{144} = -0.2014$$

$\boxed{r_{yx} = r_{xy}}$

$\therefore$ correlation is symmetric

$$r_{-3} = \frac{26}{144} = 0.181$$

As we decrease lagged k then r_k becomes +ve

see examples at page # 10, 12

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Estimation of the mean, Autocovariance and Autocorrelations.

A stationary time series is characterized by its mean μ, variance σ^2, autocorrelation e_k and ~~auto~~ partial autocorrelation ϕ_{kk}. The exact values of these parameters can be calculated if the ensemble of all possible realization is known. Otherwise they can be estimated if multiple independent realization are available. In most applications, however, it is difficult or imposible to obtain multiple realizations. Most available time series contitute only a single realization, which makes it imposible to calculate the ensemble average. For a stationary process, however, we have a natural alternative of replacing the ensemble average by the time average. In the following discussion, we examine conditions under which we can estimate with good statistical properties the mean and autocovariances and hence the autocorrelations by using time averages.

A stochastic process can be considered as a collection of random variables ordered in time. Then, a time series is a realization of a stochastic process.

i) Sample Mean

with only a single realization, a natural estimator for the mean $\mu = E(Y_t)$ of a stationary process is the sample mean.

we have a finite time series $y_1, y_2 --- y_N$ of N observations from which we can only obtain estimates of the mean μ and the autocorrelations.

Sample mean $\bar{y} = \frac{\sum_{t=1}^{N} y_t}{N}$

which is the time average of N observations. The question become whether the this estimator is a valid or good estimator. Clearly,

$$E(\bar{y}) = \frac{1}{N} \sum^{N} E(y_t)$$

$$= \frac{1}{\cancel{N}} \cancel{N}\cdot(\mu)$$

$$E(\bar{y}) = \mu$$

which implies that $\bar{y}$ is an unbiased estimator of μ.

As a measure of precision of $\bar{y}$ as an estimator of μ, we find that

$$Var(\bar{y}) = \frac{1}{N^2} \sum_{t=1}^{N} \sum_{s=1}^{N} \gamma_{t-s}$$

$$= \frac{\gamma_0}{N} \left\{ 1 + 2 \sum_{k=1}^{N-1} \left(1 - \frac{k}{N}\right) \rho_k \right\}$$

A large sample, $N \longrightarrow \infty$ approximation for this variance is given by:

$$V(\bar{y}) = \frac{\gamma_0}{N} \left\{ \overbrace{1 + 2 \sum_{k=1}^{\infty} \rho_k} \right\}$$

variation b/w time series or correlation.

Notice that the first factor in $V(\bar{y})$. $\frac{\gamma_0}{N}$ is the familiar expression for the variance of $\bar{y}$ obtained from independent random samples of size N. such that:

$$V(\bar{y}) = \frac{\sigma^2}{n}$$

$$\boxed{\gamma_0 = \sigma^2}$$

$$S.E(\bar{y}) = \frac{\sigma}{\sqrt{n}}$$

ii) Sample autocovariance.

with only a single realization, we employ the following estimators using the time average to estimate the autocovariance function C_k, i.e

$$C_k = \frac{1}{N} \sum_{t=1}^{N-k} (y_t - \bar{y})(y_{t+k} - \bar{y})$$

The sample autocovariance coefficient is at lag k. where $k = 0, 1, 2, \ldots, n$. C_k is is the estimate of the autocovariance γ_k and $\bar{y}$ is the sample mean of the time series. It can be shown that the bias in C_k is of order $1/N$.

$$r_k = \frac{c_k}{c_0}$$

The value r_k may be called the sample autocorrelation function. or (estimator).

To obtain a useful estimate of the autocorrelation function in practitice, we would typically need at least 50 observations, and the estimated autocorrelation r_k should be calculated for $k = 0, 1, \ldots n$ where k was not larger than, say $\frac{N}{4}$

iii) sample autocorrelation.

For a given observed time series $y_1, y_2, \ldots y_N$ the sample ACF (Auto Correlation Function) is defined as

$$r_k = \frac{c_k}{c_0} = \frac{\sum_{t=1}^{N-k} (y_t - \bar{y})(y_{t+k} - \bar{y})}{\sum_{t=1}^{N} (y_t - \bar{y})^2}, \quad k = 0, 1, 2, \ldots$$

where $\bar{y} = \frac{\sum_{t=1}^{N} y_t}{N}$ The sample mean of the series. A plot of r_k versus k is sometimes called a sample correlogram

Now,

It is useful to have a rough check on whether ρ_k is effectively zero beyond a certain lag, For this purpose, we can use the following expression for the approximate variance of the estimated autocorrelation coefficient of a stationary normal process given by Bartlett (1946):

$$E(r_k) = \rho_k$$

$$Var(r_k) \cong \frac{1}{N} \sum_{v=-\infty}^{\infty} \left[\rho_v^2 + \rho_{v+k}\,\rho_{v-k} - 4\rho_k\,\rho_0\,\rho_{v-k} + 2\rho_v^2\,\rho_k^2 \right]$$

For any process for which all the autocorrelation ρ_v are zero for $v > q$, all terms except the first appearing in the right hand side are zero when $k > q$. Thus for the variance of the estimated autocorrelation r_k, at lags k greater than some value q beyond which the theoretical auto correlation function may be deemed to have 'died out', for large lags Bartlett's approximation gives:

$$var(r_k) \cong \frac{1}{N}\left(1 + 2\sum_{v=1}^{q} e_v^2\right) \quad \text{(A)} \qquad k > q$$

A case of particular interest occurs for $q = 0$ that is when e_k are taken to be zero for all lags (other than lag 0), and hence the series is completely random or white noise. Then standard error from eq (A) for estimated autocorrelation r_k take the simple form.

$$Var(r_k) \cong \frac{1}{N} \quad , \quad k > 0$$

$$S.E(r_k) \cong \frac{1}{\sqrt{N}} = \sqrt{\frac{1}{N}} \quad , \quad k > 0$$

$$r_k \sim NI\left(0, \frac{1}{N}\right) \Rightarrow$$

For large N, A collection of estimated autocorrelations for diff lags

$$\bar{x} \pm 1.96\,(S.E)$$

$$0 \pm \frac{2}{\sqrt{N}}$$

$$\pm \frac{2}{\sqrt{N}}$$

Two standard error limits determined under the assumption that the series is completely random are included for the autocorrelation function.

اگر اس $\pm \frac{2}{\sqrt{N}}$ limit سے باہر ہو جائے گی تو Purely random نہیں ہوگا۔

Example

The following estimated autocorrelation were obtained from a time series of length $N=200$ generated from a stochastic process for which it was known that $\rho_1 = -0.4$ and $\rho_k = 0$ for $k \geq 2$

K	r_K
1	-0.38
2	-0.08
3	$+0.11$
4	-0.08
5	$+0.02$
6	0.00
7	0.00
8	0.00

9	0.07
10	−0.08

Solution.

on the assumption that the series is Completely random, that is white noise, we have $q=0$, then for all lags, then we have:

$$Var(r_k) \cong \frac{1}{N} = \frac{1}{200} = 0.005$$

$$S.E(r_k) = \sqrt{0.005} = 0.07$$

$$\pm \frac{2}{\sqrt{n}}$$

$$= \pm \frac{2}{\sqrt{200}}$$

$$= \pm 0.141$$

Since the value of $r_1 = -0.38$ is over five times the $S.E(r_k) = 0.07$. it Can be conclude That ρ_1 is non zero. Moreover the estimated autocorrelations for $k > 1$ are all small.

Subsituting r_1 for ρ_1 with $q=1$

~~[illegible]~~

$$Var(r_k) \simeq \frac{1}{200}\left(1+2(-0.38)^2\right)$$

$$= 0.0064 \qquad k>1$$

$$S.E(r_k) = 0.08$$

Since the estimated autocorrelations for lags greater than 1 are small compared with this standard error, there is no reason to doubt the adequacy of the model $\rho_1 \neq 0$, $\rho_k = 0$ $(k \geq 2)$

4. ⟶

Example, Past Paper.

$N = 400$

Is there any evidence of non-randomness.

K	r_K
1	0.02
2	0.05
3	−0.09
4	0.08
5	−0.02
6	0.00
7	0.12
8	0.06
9	0.02
10	−0.08

It is greater but its not means that in this correlation

one values ignored

Solution.

$$\pm \frac{1.96}{\sqrt{N}}$$

$$= \pm \frac{1.96}{\sqrt{400}} = \pm 0.098$$

$$N \times \frac{1}{N} = 1$$

It is a random process.

Properties of ACF (Auto Correlation function)

Property 1, 2 See at Page 84, 85

property 3) **chauchy - schewarz inequality:**

$$E[\underline{X}\,\underline{Y}] \leq \left[E(X^2)\,E(Y^2)\right]^{1/2}$$

$$|\gamma_k| = E\left|\{y_t - E(y_t)\}\{y_{t+k} - E(y_{t+k})\}\right|$$

$$\leq \left[E\{y_t - E(y_t)\}^2\, E\{y_{t+k} - E(y_{t+k})\}^2\right]^{1/2}$$

$$\leq \sqrt{\gamma_0\,\gamma_0}$$

$$|\gamma_k| \leq \gamma_0$$

$$|\rho_k| \leq 1$$

4) **Lack of uniqueness:**

A given stochastic process has unique covariance structure. The converse is not in general true. Thus invertibility condition is required to ensure uniqueness even for normal stationary process.

اگر ایک سے زیادہ model کے same covariance ہوں گے
تو ہم یہ جاننے کیلئے کہ ایک auto covariance کس model سے
belong کرتا ہے اسلئے ہم invertibility condition لگا دیں گے۔

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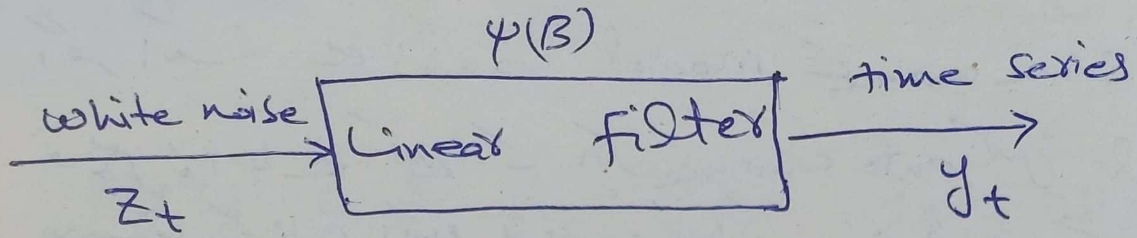
Linear filter model: (Yule, 1927)

Time series $\{Y_t\}$ in which successive values / observations are highly dependent or correlated can be regarded as generated from ~~linear~~ a series of independent "shocks" Z_t.

where Z_t is a purely random process which mean $= 0$, $var = \sigma_Z^2$

A sequence of independent random variables Z_t, Z_{t-1}, Z_{t-2} ------- is called a white noise process.

The white noise process Z_t is supposed transformed to the process Y_t is called a linear filter



The linear filtering operation simply takes a weighted sum of previous random shocks Z_t, so that

$\boxed{\psi_0 = 1}$

This Model is MA(∞) order is ∞

$$y_t = \overset{\text{parameter}}{\mu} + Z_t + \psi_1 Z_{t-1} + \psi_2 Z_{t-2} + \cdots$$

→ μ is the parameter that determines the "level" of the process.

→ $\psi_1, \psi_2, \ldots$ are weights or Parameters

→ $Z_t + \psi_1 Z_{t-1} + \psi_2 Z_{t-2} + \cdots$ is a random series.

→ $Z_t + Z_{t-1} + Z_{t-2}$ is a purely random lagged series

The above model can be written as:

$$y_t = \mu + \sum_{i=0}^{\infty} \psi_i Z_{t-i} \quad \text{— (i)}$$

From random series.

$$Z_t + \psi_1 Z_{t-1} + \psi_2 Z_{t-2} + \cdots$$

By using Backward operator

$$Z_t + \psi_1 B Z_t + \psi_2 B^2 Z_t + \cdots$$

$$(1 + \psi_1 B + \psi_2 B^2 + \cdots) Z_t$$

$$\psi(B) Z_t$$

where $\psi(B) = 1 + \psi_1 B + \psi_2 B^2 + \cdots$

put in eq (i)

$$y_t = \mu + \psi(B) Z_t$$

$\psi(B) = 1 + \psi_1 B + \psi_2 B^2 + \cdots$ is the linear operator that transforms the Z_t into y_t and is called the 'transfer function' of the filter. which converts white noise into time series.

$\psi_1, \psi_2, \text{-----}$ weights are theoritically finite or infinite. If sequence of weights is finite or infinite and absolutely summable i.e $\sum_{j=0}^{\infty} |\psi_j| < \infty$ the filter is said to be 'stable' and the process $\{y_t\}$ is stationary. The parameter μ is then the mean about which the process varies. Otherwise $\{y_t\}$ is nonstationary and μ has no specific meaning except as a reference point for the level of the process.

Moving Average, Autoregressive representation

(A) Moving Avg representation.

As we know that

$$y_t = \mu + \sum_{j=0}^{\infty} \psi_j Z_{t-j} \quad \text{—— (i)}$$

where Z_t is a zero mean and white noise. And $\sum_{j=0}^{\infty} \psi_j^2 < \infty$

By solving eq (i)

$$y_t - \mu = \sum_{j=0}^{\infty} \psi_j Z_{t-j}$$

$$\tilde{y}_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j}$$

$$\tilde{y}_t = \psi(B) Z_t \quad \text{—— (2)}$$

i) **Mean**

By using eq (i)

$$y_t = \mu + \sum_{j=0}^{\infty} \psi_j Z_{t-j}$$

$$\boxed{E(Z_{t-j}) = 0}$$

Apply Expectation.

$$E(y_t) = \mu + \sum_{j=0}^{\infty} \psi_j E(Z_{t-j})$$

$$= \mu + 0$$

$$E(y_t) = \mu$$ (mean does not depend on time)

ii) **Now, Variance**

$$Var(y_t) = Var\left[\mu + \sum_{j=0}^{\infty} \psi_j Z_{t-j}\right]$$

By definition

$$V(y_t) = E\left[y_t - E(y)\right]^2$$

$$= E\left[y_t - \mu\right]^2$$

$$= E\left[\cancel{\mu} + \sum_{j=0}^{\infty} \psi_j Z_{t-j} - \cancel{\mu}\right]^2$$

$$= E\left[\sum_{j=0}^{\infty} \psi_j Z_{t-j}\right]^2$$

$$= E\left[\sum_{j=0}^{\infty} \psi_j^2 Z_{t-j}^2 + \sum_{j}\sum_{k} \psi_j \psi_k Z_{t-j} Z_{t-k}\right]$$

$$var(y_t) = \sum_{j=0}^{\infty} \psi_j^2 E(Z_{t-j}^2)$$

$\boxed{Z_t\text{'s independent}}$

$$v(y_t) = \sigma_Z^2 \sum_{j=0}^{\infty} \psi_j^2 \quad , \; \sigma_Z^2 \geq 0$$

iii) Variance for $E\left[Z_t\, y_{t-j}\right]$

$$E\left[Z_t\, y_{t-j}\right] = \begin{cases} \sigma_Z^2 & \text{for } j=0 \\ 0 & \text{for } j>0 \end{cases}$$

$$y_t = \mu + \psi_0 Z_t + \psi_1 Z_{t-1} + \psi_2 Z_{t-2} + \cdots$$

$$y_{t-j} = \mu + \psi_0 Z_{t-j} + \psi_1 Z_{t-j-1} + \psi_2 Z_{t-j-2} + \cdots$$

$$Z_t y_{t-j} = \mu Z_t + \psi_0 Z_t Z_{t-j} + \psi_1 Z_t Z_{t-j-1} + \cdots$$

Apply Expectation.

$$E\left[Z_t\, y_{t-j}\right] = E\left[\mu Z_t + \psi_0 Z_t Z_{t-j} + \psi_1 Z_t Z_{t-j-1} + \cdots\right]$$

$$= \mu E(Z_t) + E\left[Z_t Z_{t-j}\right] + \psi_1 E\left[Z_t Z_{t-j-1}\right] + \cdots$$

$\boxed{\psi_0 = 1}$

$$E\{Z_t \, y_{t-j}\} = E(Z_t^2)$$

$$= \sigma_Z^2 \quad \text{for } j=0$$

$$= 0 \quad \text{for } j>0$$

iv) AutoCovariance

For lag 'k'

$$\gamma_k = E\{\tilde{y}_t \, \tilde{y}_{t+k}\} \longrightarrow (A)$$

where $\tilde{y}_t = \sum_{i=0}^{\infty} \psi_i Z_{t-i}$

$$\tilde{y}_{t+k} = \sum_{j=0}^{\infty} \psi_j Z_{t+k-j}$$

put the values in eq (A)

$$\gamma_k = E\left[\sum_{i=0}^{\infty} \psi_i Z_{t-i} \sum_{j=0}^{\infty} \psi_j Z_{t+k-j}\right]$$

$$= E\left\{\sum_{i=0}^{\infty} \sum_{j=0}^{\infty} \psi_i \psi_j Z_{t-i} Z_{t+k-j}\right\}$$

$$= E\Big[\sum_{i=0}^{\infty} \sum_{j=0}^{\infty} \psi_i \psi_j Z_{t-i} Z_{t+k-j}$$

$$i = j-k$$

$$+ \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} \psi_i \psi_j Z_{t-i} Z_{t+k-j} \Big]$$

$$\boxed{\begin{aligned} \cancel{t} - i &= \cancel{t} + k - j \\ -i &= k - j \\ i &= -j + k \\ i &= j - k \\ j &= i + k \end{aligned}}$$

$$\gamma_k = E\Big[\sum_{i=0}^{\infty} \psi_i \psi_{i+k} Z_{t-i}^2 \Big]$$

$$= \sum_{i=0}^{\infty} \psi_i \psi_{i+k} E(Z_{t-i}^2)$$

$$\gamma_k = \sigma_Z^2 \sum_{i=0}^{\infty} \psi_i \psi_{i+k}$$

v) Auto Correlation

$$\rho_k = \frac{\gamma_k}{\gamma_0}$$

$$= \frac{\cancel{\sigma_Z^2} \sum_{i=0}^{\infty} \psi_i \psi_{t+k}}{\cancel{\sigma_Z^2} \sum_{i=0}^{\infty} \psi_i^2}$$

$\sum_{i=0}^{\infty} \psi_i \psi_{t+k}$ is infinite series

$$\rho_k = \frac{\sum_{i=0}^{\infty} \psi_i \psi_{t+k}}{\sum_{i=0}^{\infty} \psi_i^2}$$

For stationary we have to show that 'γ_k' is finite for each lag 'k'. By using chashy shevar's inequality

$$|\gamma_k| = \left|E(\tilde{y}_t \, \tilde{y}_{t+k})\right| \leq \left[var(\tilde{y}_t)\, var(\tilde{y}_{t+k})\right]^{1/2}$$

$$E\{\tilde{y}_t \, \tilde{y}_{t+k}\} - E\{\tilde{y}_t\} E\{\tilde{y}_{t+k}\} \leq \left\{\sigma_z^2 \sum_{i=0}^{\infty} \psi_i^2 \sigma^2 \sum_{i=0}^{\infty} \psi_i^2\right\}^{\frac{1}{2}}$$

$$\leq \sigma_z^2 \sum_{i=0}^{\infty} \psi_i^2$$

$$\sum_{i=0}^{\infty} \psi_i^2 < \infty$$

Hence, $\sum_{i=0}^{\infty} \psi_i^2 < \infty$ is a required condition for the process (1) to be stationary.

B) Autoregressive Representation

$$y_t = \mu + \pi_1 y_{t-1} + \pi_2 y_{t-2} + \cdots + Z_t$$

we assume $\mu = 0$

$$y_t = \pi_1 y_{t-1} + \pi_2 y_{t-2} + \cdots + Z_t$$

$$(y_t - \pi_1 y_{t-1} - \pi_2 y_{t-2} - \cdots) = Z_t$$

By using Backward operator

$$(y_t - \pi_1 B y_t - \pi_2 B^2 y_t - \cdots) = Z_t$$

$$(1 - \pi_1 B - \pi_2 B^2 - \cdots) y_t = Z_t$$

$$\pi(B) y_t = Z_t \quad \text{—— (1)}$$

Some points to be noted

→ The process (1) be stationary the process must written in a moving Average representation.

i.e $\pi(B) y_t = Z_t$

$$y_t = [\pi(B)]^{-1} Z_t$$

$$y_t = \psi(B) Z_t$$

→ For a linear process $y_t = \psi(B) Z_t$ to be invertible so it can be written in terms of Autoregressive representation.

• ———————— •

Moving Average process (MA)

From notes at pg #28

In Moving Average representation if only a finite number of ψ weights are non-zero. i.e $\psi_1 = \theta_1, \psi_2 = \theta_2, \ldots, \psi_q = +\theta_q$ and $\psi_k = 0$ for $k > q$

$$y_t = Z_t + \theta_1 Z_{t-1} + \theta_2 Z_{t-2} + \cdots + \theta_q Z_{t-q}$$

is moving average model, MA(q).

$\theta_1, \theta_2, \ldots \theta_q, \mu, \sigma_Z^2$ are parameters.

MA(q) → order

t is the time which depends on t-1, t-2, — previous time values.

If $q = 1$ $MA(1) \Rightarrow y_t = Z_t - \theta Z_{t-1}$

Properties of MA(q) (42)

$\boxed{\theta_0 = 1}$

i) Mean

$$y_t = \theta_0 Z_t + \theta_1 Z_{t-1} + \cdots + \theta_q Z_{t-q}$$

$$y_t = Z_t + \theta_1 Z_{t-1} + \cdots + \theta_q Z_{t-q}$$

$$y_t = \sum_{i=0}^{q} \theta_i Z_{t-i}$$

time کے change ہونے سے
MA process میں
کوئی change
نہیں آئے گی۔

Apply Expectation b.s

$$E(y_t) = E\left[\sum_{i=0}^{q} \theta_i Z_{t-i}\right]$$

$$= \sum_{i=0}^{\infty} \theta_i E(Z_{t-i})$$

$$\boxed{E(Z_{t-i}) = 0} \rightarrow \mu$$

$$E(y_t) = 0$$

It does not depend on time
we assume $\mu = 0$

ii) Variance

$$V(y_t) = E(y_t^2) - [E(y_t)]^2 \quad \text{——— (i)}$$

$$E(y_t^2) = E\left[\sum_{i=0}^{q} \theta_i Z_{t-i}\right]^2$$

$$= E\left[\sum_{i=0}^{q} \theta_i^2 Z_{t-i}^2 + \sum_{i \neq j}\sum \theta_i \theta_j Z_{t-i} Z_{t-j}\right]$$

$$E(y_t^2) = \sum_{i=0}^{q} \theta_i^2 \, E(Z_{t-i}^2)$$

$$E(y_t^2) = \sigma_Z^2 \sum_{i=0}^{q} \theta_i^2$$

Put in eq (i) $\boxed{\because E(y_t) = 0}$

$$v(y_t) = \sigma_Z^2 \sum_{i=0}^{q} \theta_i^2 - (0)^2$$

$$V(y_t) = \sigma_Z^2 \sum_{i=0}^{q} \theta_i^2$$

Finite series because θ^2 is finite.

θ number of finite parameters are involved. It is not a function of time so, the variance of MA does not depend on time.

iii) AutoCovariance

The k-step backward autocovariance function of MA(q) is defined as:

$$\gamma_k = Cov(y_t, y_{t-k})$$

$$= E\left[y_t \, y_{t-k}\right] - E(y_t)\,E(y_{t-k})$$

$$\because E(y_t) = 0 = E(y_{t-k})$$

$$\gamma_k = E(y_t \, y_{t-k}) - 0$$

$$\gamma_k = E(y_t \, y_{t-k}) \quad \text{—— (A)}$$

$$\because \quad y_t = \sum_{i=0}^{q} \theta_i Z_{t-i}$$

$$y_{t-k} = \sum_{j=0}^{q} \theta_j Z_{t-k-j}$$

put in eq (A)

$$\gamma_k = E\left[\sum_{i=0}^{q} \theta_i Z_{t-i} \sum_{j=0}^{q} \theta_j Z_{t-k-j}\right]$$

$$\cancel{t} - i = \cancel{t} - k - j$$

$$-i = -k - j$$

$$j = i - k$$

$$\gamma_k = E\left[\sum_{i=0}^{q} \sum_{\substack{j=0 \\ j=i-k}}^{q} \theta_i \theta_j Z_{t-i} Z_{t-j-k} + \sum_{i=0}^{q} \sum_{\substack{j=0 \\ j \neq i-k}}^{q} \theta_i \theta_j Z_{t-i} Z_{t-j-k}\right]$$

$$\gamma_k = E\left[\sum_{i=k}^{q} \theta_i \theta_{i-k} Z_{t-i}^2 + \text{C.P. term}\right]$$

$$\gamma_k = \sum_{i=k}^{q} \theta_i \theta_{i-k} E(Z_{t-i}^2)$$

$$\gamma_k = \sigma_Z^2 \sum_{i=k}^{q} \theta_i \theta_{i-k} \quad \text{for } k = 1,2,3,\ldots q$$

. ——————————— .

If we put $k = 0$ then $\gamma_k = \gamma_0$

it becomes variance

$$\gamma_0 = \sigma_Z^2 \sum_{i=k}^{q} \theta_i^2$$

we already fined.

. ——————————— .

iv) Auto correlation (For k-step backward)

$$\rho_k = \frac{\gamma_k}{\gamma_0} \quad \text{at lag } k$$

$$= \frac{\cancel{\sigma_Z^2} \sum_{i=k}^{q} \theta_i \theta_{i-k}}{\cancel{\sigma_Z^2} \sum_{i=k}^{q} \theta_i^2}$$

$$\rho_k = \frac{\sum_{i=k}^{q} \theta_i \theta_{i-k}}{\sum_{i=k}^{q} \theta_i^2} , \quad k = 1,2,\ldots q$$

*. ——————————— *

v) Auto Covariance

The k-step forward auto covariance function of MA(q) is defined as:

$$\gamma_k = Cov(y_t \; y_{t+k})$$

$$= E(y_t \; y_{t+k}) - E(y_t) \, E(y_{t+k})$$

$$\because \; E(y_t) = 0 = E(y_{t+k})$$

$$\gamma_k = E(y_t \; y_{t+k}) - 0$$

$$\gamma_k = E(y_t \; y_{t+k}) \text{ ——— } (B)$$

$$\because \; y_t = \sum_{i=0}^{q} \theta_i Z_{t-i}$$

$$y_{t+k} = \sum_{j=0}^{q} \theta_j Z_{t+k-j}$$

Put in eq (B)

$$\gamma_k = E\left[\sum_{i=0}^{q} \theta_i Z_{t-i} \sum_{j=0}^{q} \theta_j Z_{t+k-j}\right]$$

$$\cancel{t} - i = \cancel{t} + k - j$$

$$-i = k - j$$

$$j = i + k$$

$$\gamma_k = E\left[\sum_{\substack{i=0 \\ j=i+k}}^{q} \sum_{j=0}^{q} \theta_i \theta_j Z_{t-i} Z_{t+k-j} + \sum_{\substack{i=0 \\ j\neq i+k}}^{q} \sum_{j=0}^{q} \theta_i \theta_j Z_{t-j} Z_{t+k-j}\right]$$

$$\gamma_k = E\left[\sum_{i=0}^{q-k} \theta_i \theta_{k+i} Z_{t-i}^2 + \text{C.P term}\right]$$

$$\gamma_k = \sum_{i=0}^{q-k} \theta_i \theta_{k+i} E(Z_{t-i}^2) + 0$$

$$\gamma_k = \sigma_Z^2 \sum_{i=0}^{q-k} \theta_i \theta_{k+i} \quad \text{for } k = 1, 2, 3, \ldots q$$

If we put $k=0$ then $\gamma_k = \gamma_0$

it becomes variance

$$\gamma_0 = \sigma_Z^2 \sum_{i=k}^{q} \theta_i^2$$

vi) Auto Correlation (For k-Step Forward)

$$\rho_k = \frac{\gamma_k}{\gamma_0} \quad \text{at lag } k$$

$$= \frac{\cancel{\sigma_z^2} \sum_{i=0}^{q-k} \theta_i \theta_{k+i}}{\cancel{\sigma_z^2} \sum_{i=0}^{q} \theta_i^2}$$

$$\rho_k = \frac{\sum_{i=0}^{q-k} \theta_i \theta_{k+i}}{\sum_{i=0}^{q} \theta_i^2}, \quad k = 1, 2, 3, \ldots q$$

For a stationary process

$$\rho_{-k} = \rho_k$$

correlation is symetric $r_{xy} = r_{yx}$

ρ_k (Forward), ρ_k (backward)

اگر ہم expand کریں تو دونوں situations میں
(Same) result آئے گا جو کہ indicate کر رہا ہے کہ ہم چاہے
series کو Backward، past میں یا Forward میں لیں تو دونوں
صورتوں میں ہمارے پاس same result آئے گا۔

q, order
q=1

Autocorrelation For MA(1)

The series of MA(1) is:

$$y_t = Z_t - \theta_1 Z_{t-1}$$

For $k = 1$

$$\rho_1 = \frac{\sum_{i=0}^{q-1} \theta_i \theta_{i+1}}{\sum_{i=0}^{q} \theta_i^2}$$

$$= \frac{\sum_{i=0}^{0} \theta_i \theta_{i+1}}{\sum_{i=0}^{1} \theta_i^2} = \frac{\theta_0 \theta_1}{\theta_0^2 + \theta_1^2}$$

$$\rho_1 = \frac{\theta_1}{1 + \theta_1^2}$$

Invertibility Condition of MA process

From notes at pg # 44

"This is a property that possessed by a moving average process. Invertibility solves non-uniqueness of autocorrelation function of moving average."

Autocorrelation function plays an important role in identification and estimation of a stochastic process underlying observed time series. (or model) Moving average process are always stationary but, do not have a unique autocovariance function. Non-uniquenss of ACF of MA process makes the estimation of moving average process problematic. To adress this issue we impose a condition which is called invertibility condition.

The invertibility condition is independent of the stationary condition. Invertibility is also applicable to nonstationary linear models.

For MA(1)

Consider the following first-order MA process

MA(1) A: $y_t = Z_t + \theta Z_{t-1}$

MA(1) B: $y_t = Z_t + \frac{1}{\theta} Z_{t-1}$

There two different process have exactly same a.c.f

For A

$$\rho_k = \begin{cases} 1 & k=0 \\ \frac{\theta}{1+\theta^2} & k=\pm 1 \\ 0 & k \geq 2 \end{cases}$$

For B

$$\rho_k = \begin{cases} 1 & k=0 \\ \frac{\theta}{1+\theta^2} & k=\pm 1 \\ 0 & k \geq 2 \end{cases}$$

Thus we cannot identify on 'MA' process uniquely from a given acf.

Now express models A and B by Putting Z_t in terms of Y_t, Y_{t-1} ------ By successive substitution.

$$Y_t = Z_t + \theta Z_{t-1} \quad \text{——— (1)}$$

$$Y_{t-1} = Z_{t-1} + \theta Z_{t-2} \quad \text{——— (2)}$$

$$Y_{t-2} = Z_{t-2} + \theta Z_{t-3} \quad \text{——— (3)}$$

$\vdots$

From eq (1)

$$Z_t = Y_t - \theta Z_{t-1} \quad \text{——— (A)}$$

From eq (2)

$$Z_{t-1} = Y_{t-1} - \theta Z_{t-2}$$

put this result in eq (A)

$$Z_t = Y_t - \theta\left[Y_{t-1} - \theta Z_{t-2}\right] \quad \text{——— (B)}$$

From eq (3)

$$Z_{t-2} = Y_{t-2} - \theta Z_{t-3}.$$

Put this result in eq (B)

$$Z_t = y_t - \theta y_{t-1} + \theta^2 [y_{t-2} - \theta Z_{t-3}]$$

$$Z_t = y_t - \theta y_{t-1} + \theta^2 y_{t-2} - \theta^3 Z_{t-3}$$

A: $Z_t = y_t - \theta y_{t-1} + \theta^2 y_{t-2} - \theta^3 y_{t-3} + \cdots$

B: $Z_t = y_t - \frac{1}{\theta} y_{t-1} + \frac{1}{\theta^2} y_{t-2} - \frac{1}{\theta^3} y_{t-3} + \cdots$

If $|\theta| < 1$, the series for 'A' conveges wheras that for 'B' does not. Thus if $|\theta| < 1$, model A is said to be invertible whereas model B is not. The imposition of invertibility condition ensures that there is a unique MA process for given acf.

The invertibility condition for MA(q)

$$Y_t = \theta_0 Z_t + \theta_1 Z_{t-1} + \dots + \theta_q Z_{t-q}$$

$$Y_t = \theta_0 Z_t + \theta_1 B Z_t + \dots + \theta_q B^q Z_t$$

$$Y_t = (\theta_0 + \theta_1 B + \dots + \theta_q B^q) Z_t$$

جتنا MA کا order ہو گا اتنی ہی degree کا polynomial ہو گا۔

$$Y_t = \theta(B) Z_t$$

where $\theta(B)$ is a polynomial of order 'q' in B

B: complex variable

$$\theta(B) = \theta_0 + \theta_1 B + \dots + \theta_q B^q = 0$$

MA(q) is invertible if the root of $\theta(B) = 0$ all lie outside the unit circle.

For example, for MA(1)

$$\boxed{\begin{aligned}\theta_0 &= 1\\ \theta_1 &= \theta\end{aligned}}$$

$$y_t = Z_t + \theta Z_{t-1}$$

$$y_t = (1 + \theta B) Z_t$$

$$y_t = \theta(B) Z_t$$

Put $\theta(B) = 0$

$$1 + \theta B = 0$$

$$\theta B = -1$$

$$B = -\frac{1}{\theta}$$

For invertibility $|B| > 1$

$$\left|-\frac{1}{\theta}\right| > 1$$

$$|\theta| < 1$$

Thus the root is outside the unit circle provided that $|\theta| < 1$.

Moving Avg Process (55) of order 1, MA(1)

MA(1) Model:

The moving Average process of order 1 is defined as:

$$Y_t = Z_t + \theta_1 Z_{t-1}$$

or from notes at page 28

Properties:

i) mean

$$E(y_t) = E[Z_t + \theta_1 Z_{t-1}]$$

$$= E(Z_t) + \theta_1 E(Z_{t-1})$$

$$= 0$$

It is not a function of time. Because it is not depend on time. mean is stable.

ii) Variance of MA(1) process:

$$var(y_t) = E(y_t^2) - [E(y_t)]^2$$

$$= E(y_t^2) - (0)^2 \qquad \because E(y_t) = 0$$

$$= E(y_t^2)$$

$$= E\left(Z_t + \theta_1 Z_{t-1}\right)^2$$

$$= E\left[Z_t^2 + \theta_1^2 Z_{t-1}^2 + C.P\ \text{term}\right]$$

$$= E(Z_t^2) + \theta_1^2 E(Z_{t-1}^2) + E(C.P\ \text{term})$$

$$= \sigma_Z^2 + \theta_1^2 \sigma_Z^2 + 0$$

$$Var(y_t) = \sigma_Z^2 (1 + \theta_1^2) \rightarrow \text{parameter}$$

Variance will be the same with respect to zero mean process or non zero mean process. And variance in finite. And not depend on time.

iii) AutoCovariance

The k-step backward auto covariance function of MA(1) is defined as:

$$\gamma_k = Cov(y_t, y_{t-k})$$

$$= E(y_t\ y_{t-k}) - E(y_t)E(y_{t-k})$$

$$= E(y_t\ y_{t-k}) - 0 \qquad \because E(y_t) = 0,\ E(y_{t\pm k}) = 0$$

$$\gamma_k = E(y_t \; y_{t-k})$$ (5)

$$\because \; y_t = Z_t + \theta_1 Z_{t-1}$$

$$y_{t-k} = Z_{t-k} + \theta_1 Z_{t-k-1}$$

So, $$\gamma_k = E\left[\{Z_t + \theta_1 Z_{t-1}\}\{Z_{t-k} + \theta_1 Z_{t-k-1}\}\right] \quad \text{—— (A)}$$

For $k=1$

$$\gamma_1 = E\left[\{Z_t + \theta_1 Z_{t-1}\}\{Z_{t-1} + \theta_1 Z_{t-2}\}\right]$$

we will write only square terms, all the c.p terms are vanishes.

$$\gamma_1 = E\left[\theta_1 Z_{t-1}^2 + \text{c.p terms}\right]$$

$$= \theta_1 E(Z_{t-1}^2) + E\{\text{c.p terms}\}$$

$$= \theta_1 \sigma_Z^2 + 0$$

$$\gamma_1 = \sigma_Z^2 \theta_1$$

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For $k=2$

eq (A) becomes

$$\gamma_2 = E\left[\{Z_t + \theta_1 Z_{t-1}\}\{Z_{t-2} + \theta_1 Z_{t-3}\}\right]$$

All the cross product terms vanishes.

$$\gamma_2 = E\left[\text{C.p. terms}\right]$$

$$\gamma_2 = 0$$

$$\gamma_k = \begin{cases} \sigma_Z^2(1+\theta_1^2) & k=0 \\ \sigma_Z^2\,\theta_1 & k=1 \\ 0 & k\geq 2 \end{cases}$$

iv) Auto Correlation.

$$\rho_k = \begin{cases} \dfrac{\sigma_Z^2(1+\theta_1^2)}{\sigma_Z^2(1+\theta_1^2)} & k=0 \\[2ex] \dfrac{\theta_1\,\sigma_Z^2}{\sigma_Z^2(1+\theta_1^2)} & k=1 \\[2ex] \dfrac{0}{\sigma_Z^2(1+\theta_1^2)} & k\geq 2 \end{cases}$$

$$\rho_k = \begin{cases} 1 & k = 0 \\ \dfrac{\theta_1}{1+\theta_1^2} & k = 1 \\ 0 & k \geq 2 \end{cases}$$

اگر θ +ve ہوں گے تو
+ve correlation ہوگا اور اگر
θ -ve ہوں گے -ve corr
ہوگا۔

Moving Avg process of order 2, MA(2)

MA(2) model:

The moving Average process of order 2 is defined as:

$$Y_t = Z_t + \theta_1 Z_{t-1} + \theta_2 Z_{t-2}$$

θ_1, θ_2 are MA Parameters

Z_t, Z_{t-1}, Z_{t-2} are purely random process at lagged 2 series

i) Mean

$$E(Y_t) = E\left[Z_t + \theta_1 Z_{t-1} + \theta_2 Z_{t-2}\right]$$

$$= E(Z_t) + \theta_1 E(Z_{t-1}) + \theta_2 E(Z_{t-2})$$

$$E(Y_t) = 0$$

It is not a function of time, because it is not depend on time. mean is stable.

ii) Variance

$$Var(Y_t) = E(y_t^2) - [E(y_t)]^2$$

$\because E(y_t) = 0$

$$= E(y_t^2) - 0$$

$$= E[Z_t + \theta_1 Z_{t-1} + \theta_2 Z_{t-2}]^2$$

$$= E[Z_t^2 + \theta_1^2 Z_{t-1}^2 + \theta_2^2 Z_{t-2}^2 + \text{c.p term}]$$

$$= E(Z_t^2) + \theta_1^2 E(Z_{t-1}^2) + \theta_2^2 E(Z_{t-2}^2) + E(\text{c.p terms})$$

$$= \sigma_Z^2 + \theta_1^2 \sigma_Z^2 + \theta_2^2 \sigma_Z^2 + 0$$

$$Var(Y_t) = \sigma_Z^2 (1 + \theta_1^2 + \theta_2^2)$$

variance will be the same with respect to zero mean or non zero mean process. Variance is finit and for stationary process It is not depend on time.

iii) Auto Covariance

$$\gamma_k = Cov(y_t, y_{t-k})$$

$$= E(y_t \cdot y_{t-k}) - E(y_t)\,E(y_{t-k})$$

$$= E(y_t \cdot y_{t-k}) - 0 \qquad \boxed{\because E(y_t)=0,\ E(y_{t-k})=0}$$

$$\gamma_k = E(y_t \cdot y_{t-k})$$

$$\because \quad y_t = Z_t + \theta_1 Z_{t-1} + \theta_2 Z_{t-2}$$

$$y_{t-k} = Z_{t-k} + \theta_1 Z_{t-k-1} + \theta_2 Z_{t-k-2}$$

$$\gamma_k = E\left\{(Z_t + \theta_1 Z_{t-1} + \theta_2 Z_{t-2})(Z_{t-k} + \theta_1 Z_{t-k-1} + \theta_2 Z_{t-k-2})\right\} \quad \text{——— (A)}$$

For $k=1$

$$\gamma_1 = E\left[(Z_t + \theta_1 Z_{t-1} + \theta_2 Z_{t-2})(Z_{t-1} + \theta_1 Z_{t-2} + \theta_2 Z_{t-3})\right]$$

(62)

we will write only square terms and C.P terms vanishes.

$$\gamma_1 = E\left[\theta_1 Z_{t-1}^2 + \theta_1\theta_2 Z_{t-2}^2\right]$$

$$\gamma_1 = \theta_1 E(Z_{t-1}^2) + \theta_1\theta_2 E(Z_{t-2}^2)$$

$$\gamma_1 = \theta_1 \sigma_Z^2 + \theta_1\theta_2 \sigma_Z^2$$

$$\gamma_1 = \sigma_Z^2(\theta_1 + \theta_1\theta_2)$$

$$\gamma_1 = \sigma_Z^2 \theta_1(1+\theta_2)$$ at lag $k=1$

Now, for $k = 2$

eq (A) becomes.

$$\gamma_2 = E\left\{(Z_t + \theta_1 Z_{t-1} + \theta_2 Z_{t-2})(Z_{t-2} + \theta_1 Z_{t-3} + \theta_2 Z_{t-4})\right\}$$

we will write only square terms and C.P terms will be vanished.

$$\gamma_2 = E\left[\theta_2 Z_{t-2}\right]$$

$$= \theta_2 E(Z_{t-2})$$

$$\gamma_2 = \theta_2 \sigma_Z^2$$ (63)

$$\gamma_2 = \sigma_Z^2 \theta_2 \quad \text{at lag } k=2$$

Now, For $k=3$

eq (A) becomes.

$$\gamma_3 = E\left[(Z_t + \theta_1 Z_{t-1} + \theta_2 Z_{t-2})(Z_{t-3} + \theta_1 Z_{t-4} + \theta_2 Z_{t-5})\right]$$

All the C.P terms will be vanishes.

$$\gamma_3 = E[\text{C.P terms}]$$

$$\gamma_3 = 0 \quad \text{at lag } k=3$$

$$\gamma_k = \begin{cases} \sigma_Z^2 (1+\theta_1^2+\theta_2^2) & k=0 \\ \sigma_Z^2 \theta_1 (1+\theta_2) & k=1 \\ \sigma_Z^2 \theta_2 & k=2 \\ 0 & k \geq 3 \end{cases}$$

iv) Auto Correlation.

$$\rho_k = \frac{\gamma_k}{\gamma_0}$$

$$\rho_k = \begin{cases} \dfrac{\sigma_z^2(1+\theta_1^2+\theta_2^2)}{\sigma_z^2(1+\theta_1^2+\theta_2^2)} & k=0 \\[2ex] \dfrac{\cancel{\sigma_z^2}\,\theta_1(1+\theta_2)}{\cancel{\sigma_z^2}(1+\theta_1^2+\theta_2^2)} & k=1 \\[2ex] \dfrac{\cancel{\sigma_z^2}\,\theta_2}{\cancel{\sigma_z^2}(1+\theta_1^2+\theta_2^2)} & k=2 \\[2ex] \dfrac{0}{\sigma_z^2(1+\theta_1^2+\theta_2^2)} & k\geq 3 \end{cases}$$

$$\rho_k = \begin{cases} 1 & k=0 \\ \dfrac{\theta_1(1+\theta_2)}{1+\theta_1^2+\theta_2^2} & k=1 \\ \dfrac{\theta_2}{1+\theta_1^2+\theta_2^2} & k=2 \\ 0 & k \geq 3 \end{cases}$$

From notes at page # 41

Example 7

Show that ACF (AutoCorrelation Function) of the following process

$$Y_t = \sum_{i=0}^{m} \frac{Z_{t-i}}{m+1} = \frac{1}{m+1}\left(Z_t + Z_{t-1} + \cdots + Z_{t-m}\right)$$

is given by

$$\rho_k = \begin{cases} \dfrac{m+1-k}{m+1} & , k=0,1,\cdots m \\ 0 & k \geq m+1 \end{cases}$$

Solution.

Mean

$$E(Y_t) = E\left(\sum_{i=0}^{m} \frac{Z_{t-i}}{m+1}\right)$$

$$= \frac{1}{m+1} E\left[\sum_{i=0}^{m} Z_{t-i}\right]$$

$$= \frac{1}{m+1} \sum_{i=0}^{m} E(Z_{t-i}) \qquad \boxed{\because E(Z_{t-i}) = 0}$$

$$= 0$$

Variance

$$var(Y_t) = E(Y_t^2) - \left[E(Y_t)\right]^2$$

$$= E(Y_t^2) - 0 \qquad \boxed{\because E(Y_t) = 0}$$

$$= E(Y_t^2)$$

$$= E\left[\sum_{i=0}^{m} \frac{Z_{t-i}}{m+1}\right]^2$$

$$= \frac{1}{(m+1)^2} E\left[\sum_{i=0}^{m} Z_{t-i}\right]^2$$

$$= \frac{1}{(m+1)^2} \sum_{i=0}^{m} E(Z_{t-i}^2)$$

$$V(Y_t) = \frac{1}{(m+1)^2} \sum_{i=0}^{m} \sigma_Z^2$$

$$= \frac{1}{(m+1)^{\cancel{2}}} \cancel{(m+1)} \sigma_Z^2$$

$$V(Y_t) = \frac{1}{m+1} \sigma_Z^2$$

Covariance

$$\gamma_k = Cov(y_t , y_{t-k})$$

$$= E\{y_t , y_{t-k}\} - E(y_t) \cdot E(y_{t-k})$$

$$= E\{y_t , y_{t-k}\} - 0$$

$$\gamma_k = E\{y_t , y_{t-k}\}$$

$$= E\left\{\sum_{i=0}^{m} \frac{Z_{t-i}}{m+1} , \sum_{j=0}^{m} \frac{Z_{t-k-j}}{m+1}\right\}$$

$$= \frac{1}{(m+1)^2} E\left\{\sum_{i=0}^{m} Z_{t-i} , \sum_{j=0}^{m} Z_{t-k-j}\right\}$$

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$$\gamma_k = \frac{1}{(m+1)^2} E\left[\underset{i=k+j}{\sum_{i=0}^{m}\sum_{j=0}^{m}} Z_{t-i} Z_{t-k-j} + \underset{i\neq k+j}{\sum_{i=0}^{m}\sum_{j=0}^{m}} Z_{t-i} Z_{t-k-j} \right]$$

$i = k+j$ $\qquad j=0 \Rightarrow i=k$

$j=m \Rightarrow i = k+m$ (Not possible)

$\downarrow$

$i=m$

$$\gamma_k = \frac{1}{(m+1)^2} E\left[\sum_{i=0}^{m}\sum_{j=0}^{m} Z_{t-k-j} Z_{t-k-j} + \text{cp term} \right]$$

$$= \frac{1}{(m+1)^2} E\left[\sum_{i=k}^{m} Z^2_{t-i} \right]$$

$$= \frac{1}{(m+1)^2} \sum_{i=k}^{m} E(Z^2_{t-i})$$

$$= \frac{1}{(m+1)^2} (m+1-k)\, \sigma_Z^2$$

$$\gamma_k = \frac{m+1-k}{(m+1)^2}\, \sigma_Z^2 \quad , \text{ for } k = 0, 1, \ldots m$$

$$\overbrace{\underbrace{0, 1, 2, \ldots\ldots k-1}_{k}, k, \ldots\ldots m}^{m+1}$$

$m+1-k$

Autocorrelation

$$\rho_k = \frac{\gamma_k}{\gamma_0}$$

$$= \frac{\frac{m+1-k}{(m+1)^2}\sigma_Z^2}{\frac{1}{m+1}\sigma_Z^2}$$

$$= \frac{m+1-k}{(m+1)^{\cancel{2}}} \cancel{\sigma_Z^2} \times \frac{\cancel{m+1}}{\cancel{\sigma_Z^2}}$$

$$\rho_k = \frac{m+1-k}{m+1}$$

$$\rho_k = \begin{cases} \frac{m+1-k}{m+1}, & k = 0, 1, 2, \dots, m \\ 0 & k \geq m+1 \end{cases}$$

→ See example # 3 at page # 34 from notes.

→ See example from notes at pg # 42 (Ex. 8)

Question.

Check the invertibility of the following model.

$$Y_t = Z_t - 0.5 Z_{t-1} + 0.2 Z_{t-2}$$

Solution.

$$Y_t = Z_t - 0.5 Z_{t-1} + 0.2 Z_{t-2}$$

By using Backward operator.

$$Y_t = Z_t - 0.5 B Z_t + 0.2 B^2 Z_t$$

$$Y_t = (1 - 0.5B + 0.2B^2) Z_t$$

$$\theta(B) = 1 - 0.5B + 0.2B^2$$

$a = 0.2$, $b = -0.5$, $c = 1$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$B = \frac{0.5 \pm \sqrt{(-0.5)^2 - 4(0.2)(1)}}{2(0.2)}$$

$$B = \frac{0.5 \pm \sqrt{-0.55}}{0.4}$$

$$= \frac{0.5 \pm \sqrt{0.55}\, i}{0.4}$$

$$= \frac{0.5}{0.4} \pm \frac{\sqrt{0.55}}{0.4} i$$

Distance formula $|B| = \sqrt{x^2 + y^2}$

$$= \sqrt{\left(\frac{0.5}{0.4}\right)^2 + \left(\frac{\sqrt{0.55}}{0.4}\right)^2}$$

$$|B| = 2.23 > 1$$

The given model is Invertible.

See example at Pg # 39 From notes.

Partial Autocorrelation

In addition to the autocorrelation between y_t and y_{t+k}, we may wish to calculate the correlation between y_t and y_{t+k} after their mutual linear dependency on the intervening variables $y_{t+1}, y_{t+2}, \ldots, y_{t+k-1}$ has been removed. The conditional correlation is:

$$corr(y_t, y_{t+k} / y_{t+1}, y_{t+2}, \ldots, y_{t+k-1})$$

Consider a regression model where dependent variable y_{t+k} from a zero mean stationary process is regressed on 'k' lagged variables.

$y_{t+k-1}, y_{t+k-2}, \ldots, y_{t+k-k}$

$$y_{t+k} = \phi_{k1} y_{t+k-1} + \phi_{k2} y_{t+k-2} + \cdots + \phi_{kk} y_t + Z_{t+k} \quad \text{(A)}$$

Error term

where ϕ_{ki} denote the ith regression parameter.

$$Z_{t+k} \sim IID(0, \sigma_z^2)$$

$$E(Z_{t+k}, y_{t+k-j}) = 0 \quad \text{for } j = 1, 2, \ldots, k$$

multiply by y_{t+k-j} on both sides and take Expectation on eq (A)

$$E(y_{t+k}, y_{t+k-j}) = \phi_{k1} E(y_{t+k-1} y_{t+k-j})$$

$$+ \phi_{k2} E(y_{t+k-2} y_{t+k-j}) + \cdots + \phi_{kk} E(y_t y_{t+k-j})$$

$$+ E(Z_{t+k} y_{t+k-j})$$

$$\gamma_j = \phi_{k1}\gamma_{j-1} + \phi_{k2}\gamma_{j-2} + \cdots + \phi_{kk}\gamma_{j-k}$$ 94

Dividing by variance (γ_0)

$$\rho_j = \phi_{k1}\rho_{j-1} + \phi_{k2}\rho_{j-2} + \cdots + \phi_{kk}\rho_{j-k} \qquad \boxed{j \geq 1}$$

For $j = 1, 2, \ldots\ldots k$ we have following systematic of equations.

$$\rho_1 = \phi_{k1}\rho_0 + \phi_{k2}\rho_{-1} + \cdots\cdots + \phi_{kk}\rho_{1-k} \quad \text{—— (i)}$$

$$\rho_2 = \phi_{k1}\rho_1 + \phi_{k2}\rho_0 + \cdots\cdots + \phi_{kk}\rho_{2-k} \quad \text{—— (ii)}$$

$$\vdots \qquad \vdots$$

$$\rho_k = \phi_{k1}\rho_{k-1} + \phi_{k2}\rho_{k-2} + \cdots\cdots + \phi_{kk}\rho_0$$

$\because \rho_0 = 1 \quad , \quad \rho_{-k} = \rho_k$

So,

$$\rho_1 = \phi_{k1} + \phi_k\rho_1 + \cdots\cdots + \phi_{kk}\rho_{k-1}$$

$$\rho_2 = \phi_{k1}\rho_1 + \phi_{k2} + \cdots\cdots + \phi_{kk}\rho_{k-2}$$

$$\vdots \qquad \vdots$$

$$\rho_k = \phi_{k1}\rho_{k-1} + \phi_{k2}\rho_{k-2} + \cdots\cdots + \phi_{kk}$$

In matrix form

$$\begin{bmatrix} \rho_1 \\ \rho_2 \\ \vdots \\ \rho_k \end{bmatrix} = \begin{bmatrix} 1 & \rho_1 & \cdots & \rho_{k-1} \\ \rho_1 & 1 & \cdots & \rho_{k-2} \\ \vdots & \vdots & & \vdots \\ \rho_{k-1} & \rho_{k-2} & \cdots & 1 \end{bmatrix} = \begin{bmatrix} \phi_{k1} \\ \phi_{k2} \\ \vdots \\ \phi_{kk} \end{bmatrix}$$

$$\underline{\rho} = \underset{\sim}{\rho}\, \underline{\phi}_k$$

use cramer rule successing for $k = 1, 2, \ldots$

(i) ρ_k کی ویلیو ایک ہی ویلیو میں $k=1$ کی ویلیو put کرنی ہے۔ پھر (i) اور (ii) ρ_k کی ویلیو دو ویلیوز میں put کرنی ہیں۔

$\underline{k=1}$ $\quad \rho_1 = \phi_{11}$

$\underline{k=2}$ $\quad \rho_1 = \phi_{21} + \phi_{22}\rho_1$

$$\rho_2 = \phi_{21}\rho_1 + \phi_{22}$$

$$\begin{bmatrix} \rho_1 \\ \rho_2 \end{bmatrix} = \begin{bmatrix} 1 & \rho_1 \\ \rho_1 & 1 \end{bmatrix} \begin{bmatrix} \phi_{21} \\ \phi_{22} \end{bmatrix}$$

(10)

$$\phi_{22} = \frac{\begin{vmatrix} 1 & \rho_1 \\ \rho_1 & \rho_2 \end{vmatrix}}{\begin{vmatrix} 1 & \rho_1 \\ \rho_1 & 1 \end{vmatrix}} = \frac{\rho_2 - \rho_1^2}{1 - \rho_1^2}$$

For generilization.

$$\phi_{kk} = \frac{\begin{vmatrix} 1 & \rho_1 & \cdots & \rho_1 \\ \rho_1 & 1 & \cdots & \rho_2 \\ \vdots & \vdots & & \vdots \\ \rho_{k-1} & \rho_{k-2} & \cdots & \rho_k \end{vmatrix}}{\begin{vmatrix} 1 & \rho_1 & \cdots & \rho_{k-1} \\ \rho_1 & 1 & \cdots & \rho_{k-2} \\ \vdots & \vdots & & \vdots \\ \rho_{k-1} & \rho_{k-2} & \cdots & 1 \end{vmatrix}}$$

Partial Autocorrelation For MA(1)

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MA(1) model is

$$y_t = Z_t + \theta Z_{t-1}$$

$$\rho_k = \begin{cases} \dfrac{\theta_1}{1+\theta_1^2} & , k=1 \\ 0 & , k>1 \end{cases}$$

Partial Autocorrelation is

$k=1$

$$\phi_{11} = \rho_1$$

$$\phi_{11} = \frac{\theta_1}{1+\theta_1^2}$$

multiply and divide with $(1-\theta_1^2)$

$$\phi_{11} = \frac{1-\theta_1^2}{1-\theta_1^2} \frac{\theta_1}{1+\theta_1^2}$$

$$= \frac{\theta_1(1-\theta_1^2)}{1-\theta_1^4}$$

$$\phi_{11} = \frac{\theta_1(1-\theta_1^2)}{1-\theta_1^4}$$

For $k=2$

$$\phi_{22} = \frac{\rho_2 - \rho_1^2}{1-\rho_1^2}$$

$\boxed{\rho_2 = 0 \text{ for } k>1}$

$$\phi_{22} = \frac{0-\rho_1^2}{1-\rho_1^2}$$

put the value of $\rho_1 = \dfrac{\theta_1}{1+\theta_1^2}$

$$\phi_{22} = \frac{\dfrac{-\theta_1^2}{(1+\theta_1^2)^2}}{1-\dfrac{\theta_1^2}{(1+\theta_1^2)^2}}$$

$$= \frac{\dfrac{-\theta_1^2}{\cancel{(1+\theta_1^2)^2}}}{\dfrac{(1+\theta_1^2)^2 - \theta_1^2}{\cancel{(1+\theta_1^2)^2}}}$$

$$= \frac{-\theta_1^2}{(1+\theta_1^2)^2 - \theta_1^2}$$

$$\phi_{22} = \frac{-\theta_1^2}{1+\theta_1^4+2\theta_1^2-\theta_1^2}$$

$$= \frac{-\theta_1^2}{1+\theta_1^2+\theta_1^4}$$

$$\phi_{22} = \frac{-\theta_1^2(1-\theta_1^2)}{1-\theta_1^6} \quad \text{at } k=2$$

$$\boxed{\phi_{33} = \frac{-\theta_1^3(1-\theta_1^2)}{(1-\theta_1^8)}}$$

For MA(1) $\Rightarrow$ $|\rho_1| < 0.5$

$$|\phi_{kk}| < 0.5$$